# DISCRETIO Dataset and Pipeline Plan

## What the project is about

DISCRETIO is a benchmark idea for world models. The question is simple. When a world model predicts what happens next in a traffic scene, does it carry hidden assumptions about how people behave, and are those assumptions close to what really happens?

We test this by changing one thing in a real scene at a time and watching how the model's predictions shift. The things we change fall into four groups.

- Physical, for example how long a pedestrian is hidden behind something
- Behavioral, for example the age or vulnerability of a pedestrian
- Social, for example the role of a road user such as a delivery rider or an ordinary commuter
- Contextual, for example the type of street or environment

To judge whether a model's shift is reasonable, we need a real world reference. We call this the Distributional Ground Truth. It is simply the observed spread of outcomes across many real situations that share the same conditions. For example, among all real cases of a child approaching an unsignalized crossing at a similar speed, how often did they cross, wait, or hesitate.

The important point is that this reference must come from real data. If we generate it with another model, we are only comparing one model's assumptions with another model's assumptions, and reviewers will notice.

## Notes

- We will not use a video generation model or a simulator to produce ground truth.
- I think FactorJEPA's DenseWorld 115K dataset should be dropped since better matched datasets exist for what we need.
- We will not pool samples across datasets unless it adds real value. More on this below.

## Candidate datasets

### IDD-PeD

- Indian pedestrian dataset from IIIT Hyderabad, released in 2025

- Unstructured traffic with unsignalized junctions, jaywalking, heavy occlusion, changing light, and pedestrians who start crossing and then stop
- Around 5000 pedestrians and roughly 205 thousand annotated frames
- Has bounding boxes for pedestrians and nearby objects with occlusion flags
- Has behavior tags such as crossing versus jaywalking, hand gestures, weaving through traffic, walking versus sprinting, looking over the shoulder, distraction, group type, and communicating with other road users
- Has scene tags such as road type, intersection type, signal or crosswalk type, and time of day
- Has an interaction flag between pedestrian and the ego vehicle
- Has ego vehicle speed and GPS at ten hertz, plus 2D pose
- The paper reports that models trained on Western and Japanese datasets drop sharply on it, which supports our contextual prior idea
- Best use for us is the unstructured side of the contextual comparison and the main source for Indian behavior statistics

## JAAD

- Roughly 346 clips taken from about 240 hours of driving video in North America and Eastern Europe
- Each pedestrian has demographic tags such as age and gender, plus behavior tags and occlusion flags, plus scene context like crosswalk type and signals
- Best use for us is reading age conditioned crossing behavior almost directly from the labels

## PIE

- Toronto driving footage, about six continuous hours, 55 sequences, around 293 thousand frames, and about 1800 pedestrians
- Has crossing intention labels and ego vehicle information
- Continuous footage means base rates are less biased toward only interesting moments
- Best use for us is a cleaner base rate for crossing behavior in a structured setting

## TITAN

- About 700 clips from busy Tokyo traffic
- Around 50 labels including pedestrian age groups and a layered set of actions such as simple actions, contextual actions, and communicative actions
- Best use for us is the closest existing thing to social behavior labels, and another age labeled structured reference

### inD, rounD, highD, exiD family

- Drone recordings of fixed intersections and roundabouts from RWTH Aachen
- Very accurate positions and the same site recorded for many hours
- Large sample sizes for the same geometry, which gives stable statistics
- rounD includes negotiation between pedestrians, motorcycles, and bicycles
- Does not come with age or role labels
- Best use for us is physical and general social negotiation statistics where sample size matters

### Quick summary of which dataset serves which group

| Prior group | Main dataset | Supporting dataset |
|---|---|---|
| Physical | IDD-PeD occlusion flags | inD and rounD, PIE |
| Behavioral | IDD-PeD, JAAD, TITAN | PIE |
| Social | IDD-PeD vehicle crops with a role classifier we add | rounD, TITAN |
| Contextual | IDD-PeD against JAAD and PIE | TITAN |

## What no dataset gives us

Nobody labels role identity, for example a food delivery rider versus a normal commuter. We will need to add this ourselves. The plan is to run an off the shelf vision language model on vehicle and rider crops that IDD-PeD already tracks, then have a human check a sample. This is much smaller than labeling raw video.

IDD-PeD may also lack a general age label for single pedestrians. The paper mentions a child with adult group tag but not clearly a general age field. We need to check the actual released files. If it is missing, we add an age bracket with a pretrained classifier and check a sample by hand.

# How to proceed after picking the datasets

The idea is to build buckets. A bucket is a group of real cases that share the same held constant conditions and the same level of the one thing we care about. Inside each bucket we count outcomes. That count is our ground truth.

## Step 1. Flatten the data

Join everything into one table with one row per pedestrian per frame. Include the box, occlusion flag, behavior tags, scene tags, interaction flag, ego speed, GPS, and pose.

## Step 2. Add movement features

For each pedestrian track, compute speed, heading, and time to the crossing point or time to collision using box movement and ego speed.

## Step 3. Fill the two missing labels

Add age bracket and role type using classifiers, and check a sample by hand. // gotta check if checking manually is actually needed or not

## Step 4. Define buckets for each prior group

Physical

- Take every track with partial or full occlusion
- Bucket by occlusion length, speed before occlusion, and signal type
- Outcome is whether the pedestrian reappears and how far the reappearance is from a straight line guess based on earlier motion
- Watch out for tracks that end because the pedestrian left the camera view and not because of real occlusion

Behavioral

- Bucket by speed and time to collision, signal type, interaction flag, and age bracket
- Outcome is how often they cross, jaywalk, or do not cross, plus how often they hesitate

Social

- Bucket by role type and by matched interaction geometry such as approach angle, gap size, and relative speed
- Outcome is who yields, how much speed changes, and which gap is accepted

Contextual

- Bucket by signal type, road type, time of day, and movement bin
- Outcome is the spread of cross, jaywalk, and wait
- For the structured against unstructured comparison, match IDD-PeD cases with no signal against JAAD or PIE cases with crosswalk and signal at similar speed and time to collision

### Step 5. Handle small buckets

Set a minimum count per bucket up front, for example thirty cases. If a bucket is smaller, merge to a coarser bucket by dropping the finest movement split first. Keep the age, role, and context splits because those are what the paper is about.

### Step 6. Store the result as a lookup table

The output is a table that maps each bucket to its outcome distribution. This is the ground truth the calibration score uses. The calibration score is one minus the Jensen Shannon divergence between the model's predicted distribution and the real one.

### Step 7. Build the counterfactual probes

This is separate from ground truth. For each real scene, create a version where one factor is changed while movement, speed, heading, and scene layout stay the same. These are the videos we show to the models.

### Step 8. Run the models and score

Run the chosen baseline world models on the original and changed scenes. Measure how much the prediction shifts and how well the shift matches the real distribution. Report per prior group and keep the structured and unstructured contexts separate.

## About pooling datasets

Merging samples is fine only when we are not merging across the thing we want to measure. Combining JAAD, PIE, and TITAN into one structured group is reasonable if we need more samples because they are the same kind of setting. Merging any of them with IDD-PeD is not fine, because the structured against unstructured difference is exactly what the contextual group measures. Keep those as separate reference distributions and separate columns in the results.

## Open questions we need to find out by experiment

- Does the released IDD-PeD data really lack a general age label, and if so how accurate is an added age classifier on these crowded and often blurry crops
- How well does a vision language model spot delivery riders and other roles in Indian traffic, and how many hand checked examples do we need
- Are there enough cases per bucket after we match on movement, or do we end up merging too much
- Is the match between IDD-PeD and JAAD or PIE close enough to call the contextual comparison fair, given different cameras, frame rates, and annotation habits
- What is the best way to make counterfactual videos that look real and keep movement fixed
- Which baseline models can we actually run with the compute we have

# Suggested staging

Since the full plan is large, a safer path is to finish one prior group end to end first. The behavioral group needs the least new work because JAAD, PIE, TITAN, and IDD-PeD already carry most of the labels. That gives a working pipeline and a real result early, and the other groups can follow.